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SVPWM Based Active Filtering Control of Dual Inverter Fed Open-end Winding Induction Motor Drive for Harmonic Mitigation

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Abstract

This research presents a open - end winding induction motor drive for harmonic mitigation using active filtering control. Two independent voltage source inverters power the open stator windings in open ended winding motor drives. When two two-level inverters are combined with half the DC link voltage, a three-level inverter is produced. Normally the two inverters share the power requirement of motor equally. In the proposed technique, a primary two-level inverter controlled by space vector pulse width modulation powers the motor from one of the open ends. The stator windings' other end is supplied by a secondary inverter equipped with a floating capacitor. To reduce phase voltage harmonic distortions, the secondary inverter is run using a series active power filter control technique. The filter scheme's reference voltage is determined by extracting the fundamental component of the primary inverter's output voltage. Comparing the proposed approach to the traditional three-level inverter implemented with an open-end winding induction motor, simulation and experimental findings demonstrate a considerable reduction in phase voltage and current harmonic distortions. This enhancement is made without the need for any additional hardware.

Keywords: Active filters, Harmonic analysis, Induction motor, Multilevel inverter, Open-End Winding Induction Motor, Space Vector Pulse Width Modulation

Conflicts of interest/Competing interests

All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

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Nomenclature

PCC	Point of Common Coupling
SAF	Series Active Filter
DVR	Dynamic Voltage Restorer

OEWM	Open-end Winding Induction Motor
SVPWM	Space Vector Pulse Width Modulation
NPC	Neutral Point Clamped
VSI	Voltage Source Inverter

1. Introduction

The introduction of semiconductor-based power converters in the commercial power sector has enhanced industry performance through energy savings, reduced maintenance, decreased downtime, lowered running costs, and increased production. However, these converters also introduce challenges such as harmonic distortion, low power factor, non-sinusoidal supply voltages (including voltage dips, swells, unbalances, and flicker), reactive power loads, and reduced system efficiency [1]. High-power industrial loads, such as variable speed drives, uninterruptible power supplies, and thyristor converters, as well as low-power home and commercial loads like televisions, fax machines, computers, fluorescent tubes, and ovens, draw distorted currents, leading to non-sinusoidal power supply voltages.

In the past, harmonics were reduced using passive LC filters. However, they have disadvantages including resonance problems, bulky, and fixed offset [2]. Active power filters, sometimes referred to as active power line conditioners, were developed to solve these issues. Active filters effectively counteract the non-sinusoidal demand of non-linear loads by introducing currents that are opposite in phase but same in magnitude at the Point of Common Coupling (PCC) [3][4]. However, in reality, the objective is to minimize harmonic distortion to a level that is tolerable under particular circumstances. Active power filters are still constrained by their capacity and cost. Hybrid filters efficiently lower the size and cost of active filters by combining several active and passive filters. A shunt filter is recommended in [5] to reduce the adverse effects that could impair the capacitor's lifespan by balancing the dc-link capacitor. To provide features like reducing inverter losses, balancing source currents, improving power factor and regulating voltage, a control scheme for DSTATCOM is suggested in [6]. To address problems including voltage sags, swells, harmonics, and unbalanced voltages in electrical systems, power quality enhancement tool like Series Active Filter (SAF) can be employed. In order to keep a clean and steady voltage supply, it works by injecting compensating voltage in series with the load [7]. The series filter regulates the amplitude of the fundamental current by generating an error signal between the load voltage and a preset reference. For a three-phase, three-wire series active power filter, a control method based on generalized p-q theory is suggested to manage the reactive power and harmonics produced by non-linear loads [8]. A Dynamic Voltage Restorer (DVR) control mechanism is proposed in [9] to rectify load voltage variations and provide a smoother and more stable voltage profile with lower harmonic content. In [10], a rotating d-q reference frame controller is used to operate a photo voltaic-fed multilevel inverter based DVR. The suggested approach maintains a steady DC connection voltage while effectively reducing voltage sag and uses asymmetrical DC sources to produce approximate sinusoidal voltage.

Harmonic distortion in inverter-fed induction machines is a significant concern in variable-speed drive applications, where voltage-source inverters supply machines with variable voltage and frequency (V/f) to control speed and torque. Unlike harmonics originating from non-linear loads on the grid, these harmonics arise primarily from the switching actions of the power electronic converter and the interaction between the inverter and the machine. High harmonic content can lead

to increased current distortion, torque ripple, acoustic noise, and reduced machine efficiency, which can degrade overall drive performance. Therefore, mitigation of these harmonics is essential to improve electromagnetic performance, reduce mechanical stress, and ensure reliable operation of the machine.

In this context, Optimal Excitation Wind Induction Machines (OEWIMs) require effective control and converter strategies to minimize harmonic content while maintaining desired torque and speed characteristics. The present work focuses on developing a modified power supply and control scheme that reduces both low- and medium-order harmonics in the stator currents, thereby improving torque quality and overall performance without modifying the machine itself. This approach provides a practical solution for achieving high-performance variable-speed operation with minimal harmonic distortion, addressing challenges specific to inverter-fed OEWIM drives.

Multilevel inverters provide useful solutions for variable frequency drives in medium-voltage and high-power sectors. In addition to offering other advantages like lower switching losses, less electromagnetic interference, low common-mode voltage, and decreased device stress [11]. The principal drawback of a multilevel inverters is that it requires more switching devices and capacitors, as well as isolated DC voltage sources. A multilevel inverter topology based on a twin inverter supplied Open-end Winding Induction Motor (OEWIM) is proposed in [12]. By using the dual inverter fed OEWIM, a three-level inverter is created with each voltage source inverter receiving half of the DC-link voltage as inputs. Compared to traditional wye or delta-connected machines, open-end winding drives have many advantages. These include lower DC-link voltage needs, a stable, fluctuation-free neutral point voltage, flexibility in voltage levels, higher power output, fault-tolerant operation, lower harmonics, and a decreased switching common mode voltage. A power circuit design for three-level inversion is presented in [13]. Two two-level inverters are connected in a cascade to achieve three-level inversion. Each inverter is powered by an independent DC power supply. By clamping one inverter to zero and switching the other, the circuit can be utilized as a two-level inverter with less modulation.

Space Vector Pulse Width Modulation (SVPWM) outperforms other PWM algorithms in terms of digital implementation, better utilization of the DC bus voltage, reduced switching losses and minimized harmonic distortions. SVPWM, as opposed to conventional sinusoidal PWM views the inverter's output as a single space vector that is manipulability in a two-dimensional field. In order to improve DC bus voltage usage and reduce harmonic distortion, SVPWM efficiently uses all possible switching states by encoding the three-phase voltages as vectors [14]. A control strategy is introduced in [15] to mitigate phase voltage distortion specifically tailored for open-end winding configurations in medium voltage AC motor drive applications. The machine receives active power from a multilevel main inverter on one side, while a two-level auxiliary inverter on the other side functions as an active power filter. In [16], a three-phase induction motor drive with open stator phase windings and a dual-bridge inverter driven by a single DC voltage source are advised. A distinct SVPWM topology employing a decoupled algorithm is utilized for voltage distortion control.

Without changing the device's voltage rating or power supply circuitry, the enhanced SVPWM system outlined in [17] provides better harmonic performance than the decoupled SVPWM method. In order to provide frequency-independent AC and DC link loads with regulated power simultaneously, an induction generator system with open stator-windings is suggested [18]. In this setup, a capacitor-driven

Voltage Source Inverter (VSI) supplies the necessary reactive power to the machine maintaining both DC link voltage and AC load voltage.

Neutral Point Clamped (NPC) inverters are commonly employed in medium and high-power applications such as renewable energy systems, industrial drives, and electric vehicles, where improved power quality and efficiency are critical. Depending on the architecture, the NPC inverter usually works by dividing the DC voltage into multiple levels. Reduced harmonic distortion, enhanced power quality, and decreased switching losses are the primary benefits of NPC multilevel inverters, which makes them appropriate for high power and efficiency applications like industrial drives, renewable energy systems, and HVDC (High Voltage Direct Current) systems. A typical NPC multilevel inverter has 27 switching states spread across 19 space vector locations [19-21]. In contrast, the OEWIM configuration [22-25] offers a greater number of switch combinations across the same 19 space vector positions, demonstrating its superiority over the NPC three-level inverter regarding flexibility and control capability.

In order to reduce harmonics, this paper proposes a series active filter technique for OEWIM drives. Traditional three-level inverter systems for OEWIM use two separate two-level inverters. Each VSI supplies half of the DC-link voltage to the motor's opposing sides. The suggested method makes use of a secondary two-level VSI with a series filter control scheme at one end of the motor and a primary two-level VSI that runs with SVPWM at the other end. The output voltage of this secondary inverter is dynamically adjusted to offset the harmonic distortions caused by the first inverter. A voltage that removes the harmonics of the motor phases is produced by the secondary voltage source inverter. The suggested drive scheme has a lower overall harmonic distortion than a traditional three-level open-ended motor, and it also has the benefit of not requiring additional switching devices.

2. Three Level Inverter Using Open-End Winding Induction Motor

Fig. 1 shows the schematic diagram of a three-level inverter realized by OEWIM. An induction motor is referred to as an open winding motor when its stator windings are open. This configuration is achieved by first opening the stator winding's neutral point which was initially star-connected and then feeding each side through two different voltage source inverters. Each inverter has its own isolated DC source, often using isolated transformer which allows complete electrical isolation, reducing common-mode voltages and improving electromagnetic compatibility. But improper grounding of inverters can cause floating references and noise injection. In the basic configuration of OEWIM, the required active power for the motor is split between two drives, typically of similar size, each powered by separate power supplies. The AC motor receives the same amount of active power from both inverters. Each inverter individually generates eight switching states, resulting in 64 total switching combinations distributed across 19 positions in the 3-level space vector diagram.

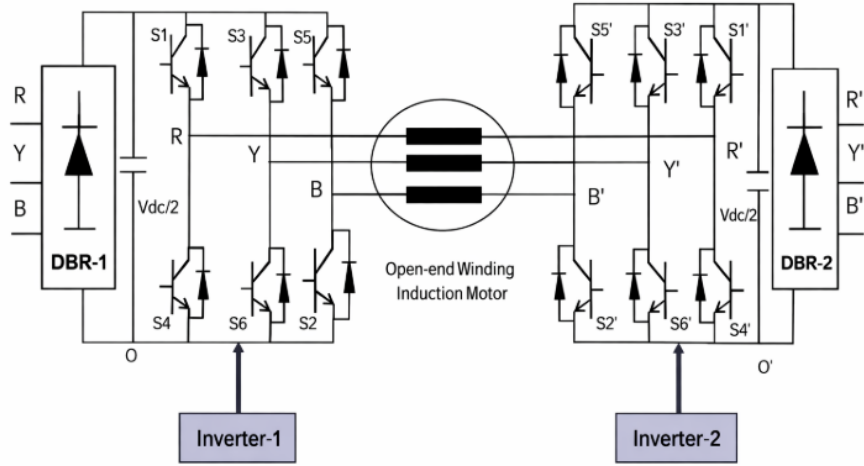


Fig. 1. Schematic diagram of Open - End Winding Induction Motor drive

The pole voltages are represented by the voltages between the DC neutral and each phase of the Inverter-1 $[V_{RO}, V_{YO}, V_{BO}]$ and inverter-2 $[V_{R'O'}, V_{Y'O'}, V_{B'O'}]$ respectively. The space phasor diagrams for both Inverters are shown in Fig. 2. The Inverter-1 switching states are indicated by numbers 1-8, where "1" means the higher device is ON and "0" means the lower device is ON. According to the same convention, the Inverter-2 switching states are represented by the numerals 1'-8'.

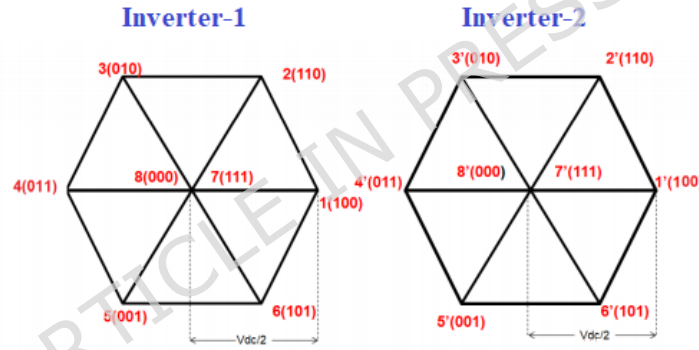


Fig. 2. Space-vector diagram of dual inverters

The resulting space-phasor diagram, shown in Fig. 3, illustrates that the two-level VSI DC-link voltage, represented by $|OA|$, is equal to $V_{dc}/2$, while the three-level inverter DC-link voltage, represented by $|OG|$, is equal to V_{dc} . The output voltage level $V_{RR'}$, or the effective motor phase voltage, is determined by the following formula, if V_{RO} and $V_{R'O'}$ stand for the pole voltages of Inverter-1 and Inverter-2, respectively:

$$V_{RR} \square V_{RO} \square V_{R'O'} \quad (1)$$

$$V_{YY'} \square V_{YO} \square V_{Y'O'} \quad (2)$$

$$V_{BB} \square V_{BO} \square V_{B'O'} \quad (3)$$

These equations explain the connections between the motor's RMS phase voltages and the pole voltages of Inverter -1 and Inverter-2.

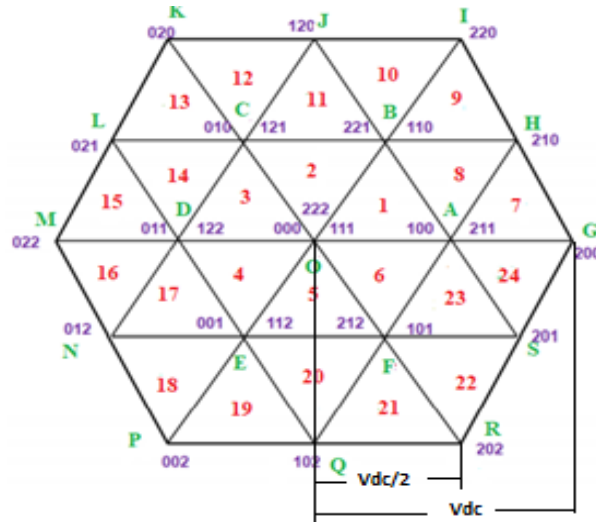


Fig. 3. Three-level space vector diagram

3. Proposed OEWM with Series Compensation

The circuit diagram for the proposed three-level OEWM drive system is shown in Fig. 4 with SVPWM controller for primary inverter and active filter controller for the secondary inverter. This study examines an OEWM configuration in which a secondary two-level inverter is used on one end of the motor and a primary inverter on the other end. The secondary inverter functions as an active filter, while the primary inverter provides all active power requirements. With this setup, there is no need for an additional independent power source because it has zero average active power and may be fueled by a stray capacitor. The primary inverter is driven using highly efficient space vector pulse width modulation (SVPWM), while the secondary unit functions as an active series filter to mitigate harmonic distortion. This approach aims to minimize harmonic components in the phase voltage, thereby reducing torque ripple and enhancing overall drive efficiency. In dual-inverter OEWM system, both inverters may experience ripple, either independently or together. But high frequency SVPWM scheme suppresses the ripples.

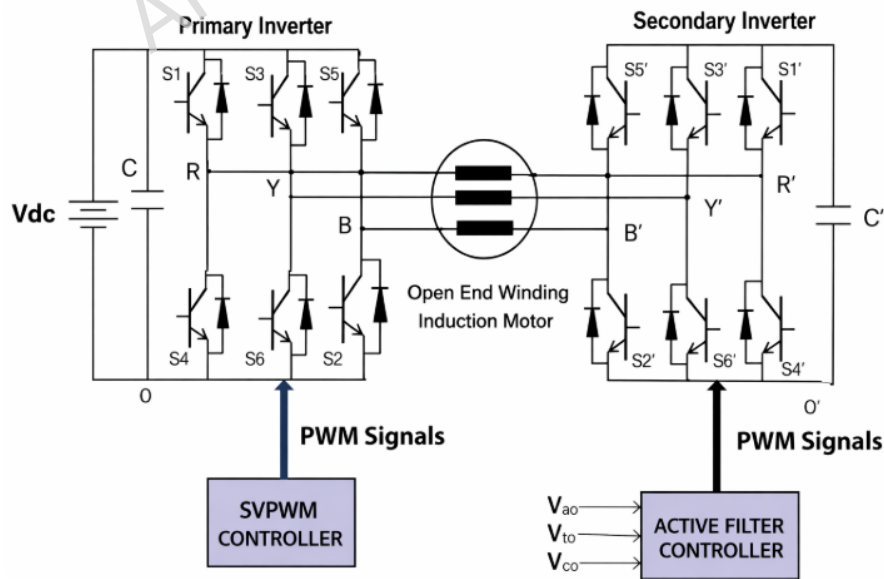


Fig. 4. Proposed OEWM Scheme

The control algorithm for the secondary inverter ensures that it generates an output voltage matching the harmonic distortion of the two-level primary inverter. The motor phase voltage generated by the OEWM is the difference between the voltages of the primary and secondary inverters. Thus, the proposed method efficiently decreases the amount of harmonics of the motor voltage.

3.1 Primary Inverter

The open winding motor has a two-level primary inverter at one end that operates at a high frequency using space vector pulse width modulation. The idea behind SVPWM is to use discrete switching states of an inverter to create sinusoidal waveforms and to represent three-phase voltage vectors in a two-dimensional coordinate system (α - β plane). To produce the required output voltage, SVPWM makes use of the idea of a spinning space vector. It uses discrete switching states of an inverter to approximate sinusoidal waveforms. The SVPWM technique put forth in [14] functions according to valid time, which is the amount of time the load is connected to the power source. This technique eliminates the requirement for reference space vectors and does not rely on lookup tables for size or angle information.

3.2 Secondary Inverter

The secondary inverter on the other end of the motor functions as an active series filter to minimize distortion in motor phase voltages with a floating capacitor with zero active power supply to the motor. Acting akin to a dynamic voltage restorer, the active series filter works to reduce harmonic distortions in the voltage waveforms. By being positioned in series with the load, an active series filter effectively addresses voltage imbalances and mitigates harmonic distortions. The series filter control method aims to deliver clean, balanced sinusoidal voltages to each motor phase. Let, U_M represents the output voltage of each phase of the induction motor, while U_P and U_S denote the output voltages of the primary two-level inverter and the secondary inverter, respectively. Therefore, the vector sum of the two-level primary-inverter voltage U_P and the inverted secondary inverter voltage U_S equals the motor voltage U_M with both open ends [26, 27]. The fundamental component U_F and harmonic component U_H of the primary inverter's output U_P can then be isolated. It is assumed that the system is balanced so that odd harmonic voltages, rather than triples, make up the voltage U_H .

$$U_M = U_P + U_S \quad (4)$$

$$U_P = U_F + U_H \quad (5)$$

$$U_H = \sum_k^{5,7,11,13} U_k \quad (6)$$

$$U_S = U_P - U_F - U_H \quad (7)$$

The basic operation of the active filter control is shown in Fig. 5. The primary level inverter output voltages V_{RO} , V_{YO} and V_{BO} are filtered with a 50 Hz cutoff frequency using a biquad low-frequency filter. The output is a 90° displaced fundamental voltage represented by the symbols, V_R , V_Y and V_B . To get the first order voltage with phase shift removed, represented as V_{R1} , V_{Y1} and V_{B1} a phase shifter block is utilized. The output of the phase shifter is divided by the peak value acquired from a peak detector in order to obtain the unit amplitude fundamental component of the inverter voltages as U_{R1} , U_{Y1} and U_{B1} . The rated load voltage magnitude is then multiplied by the unit magnitude fundamental voltage to obtain the desired inverter voltages, V_{d_R} , V_{d_Y} and V_{d_B} . The necessary voltage is subtracted from the actual value to determine the filter reference voltage. A SVPWM controller is used to determine the trigger signals for the secondary inverter switches based on the filter reference voltages V_{ref_R} , V_{ref_Y} and V_{ref_B} .

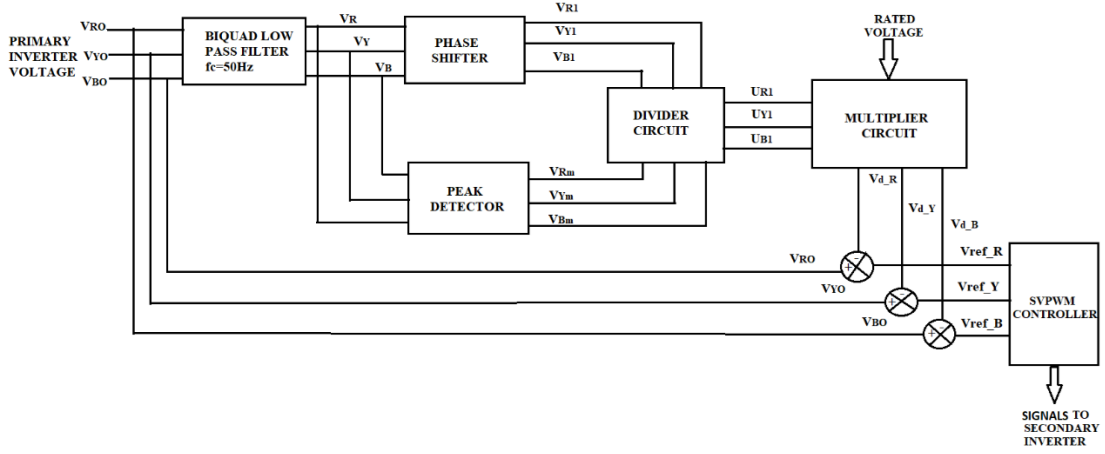


Fig. 5. Proposed series filter control method

The primary inverter output pole voltages are given as:

$$V_{RO} = V_{Rm} \sin(\omega t) \quad (8)$$

$$V_{YO} = V_{Ym} \sin(\omega t + 120^\circ) \quad (9)$$

$$V_{BO} = V_{Bm} \sin(\omega t + 120^\circ) \quad (10)$$

The biquad filter output is a fundamental component of inverter voltages with 90° phase shift as follows:

$$V_R = V_{Rm} \sin(\omega t + 90^\circ) \quad (11)$$

$$V_Y = V_{Ym} \sin(\omega t + 210^\circ) \quad (12)$$

$$V_B = V_{Bm} \sin(\omega t + 30^\circ) \quad (13)$$

The phase shifter blocks eliminate the 90° phase shift from the filter output as below:

$$V_{R1} = V_{Rm} \sin(\omega t) \quad (14)$$

$$V_{Y1} = V_{Ym} \sin(\omega t + 120^\circ) \quad (15)$$

$$V_{B1} = V_{Bm} \sin(\omega t + 120^\circ) \quad (16)$$

Unit amplitude fundamental components of the inverter voltages are as U_{R1} , U_{Y1} and U_{B1} :

$$U_{R1} = V_{R1} / V_{Rm} \quad (17)$$

$$U_{Y1} = V_{Y1} / V_{Ym} \quad (18)$$

$$U_{B1} = V_{B1} / V_{Bm} \quad (19)$$

The desired output voltage for the primary inverter is the product of unit amplitude fundamental component and rated voltage amplitude V_m :

$$V_{dR} = U_{R1} \cdot V_m \quad (20)$$

$$V_{dY} = U_{Y1} \cdot V_m \quad (21)$$

$$V_{dB} = U_{B1} \cdot V_m \quad (22)$$

Where

$$V_m = V_L = \sqrt{2} \quad (23)$$

$$V_L = \frac{V_{R1} + V_{Y1} + V_{B1}}{3} \quad (24)$$

The difference between the main inverter's output and the desired output voltage is the secondary inverter's reference voltage:

$$V_{ref_R} = V_{RO} - V_{d_R} \quad (25)$$

$$V_{ref_Y} = V_{YO} - V_{d_Y} \quad (26)$$

$$V_{ref_B} = V_{BO} - V_{d_B} \quad (27)$$

4 Results and Discussion

The MATLAB Simulink is utilized to model the recommended OEWM scheme for an open-loop, 5 HP, 3 Φ induction motor with 1 kHz switching frequency for the controller. The suggested scheme is simulated and compared with the traditional open-ended induction motor based three-level inverter. The proposed technique reduces output voltage distortion while using the same number of switches as the standard strategy. The efficacy of the plan is confirmed by the motor phase voltage. Fig. 6 displays the R phase output voltage with respect to DC link neutral point of the primary and second inverters as well as the phase voltage of the three-level OEWM drive. A THD of 12.2% is revealed by the harmonic spectrum of the induction motor phase voltage.

The proposed OEWM scheme with series active filter control is also simulated in MATLAB Simulink. The gating pulses for secondary inverter, actual filter output, and motor output phase are shown in Fig. 7. The proposed OEWM configuration reduces the overall harmonic distortion of the induction motor phase voltage to 4.6% which is well within the IEEE standards. The high frequency noise in the motor voltage waveform is due to the fast-switching action of inverters which can be mitigated by synchronous switching/interleaving. The proposed scheme and OEWM configurations based on individual harmonic distortion in relation to percentage fundamental and Total Harmonic Distortion (THD) are shown in Fig. 8. The graph confirms that, in comparison to the traditional method, the proposed OEWM arrangement produces lesser harmonic distortions.

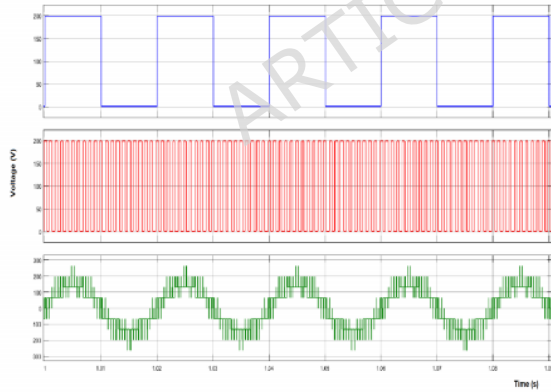


Fig. 6. Plots 1 and 2 are the pole-voltage of the inverter 1 and 2, respectively; Plot 3 shows the phase voltage of the three-level OEWM drive for $m = 0.8$

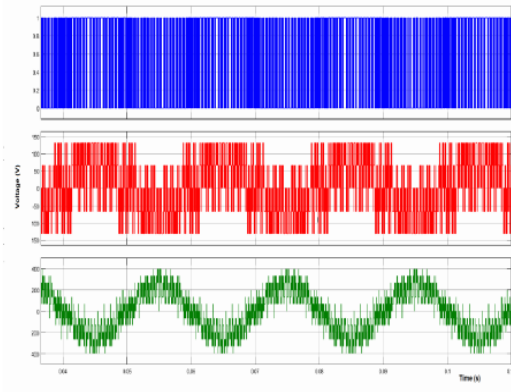


Fig. 7. Plot-1 shows the trigger signals for the secondary inverter switch; Plot-2 shows the actual filter output voltage; and Plot-3 shows the phase voltage of the motor

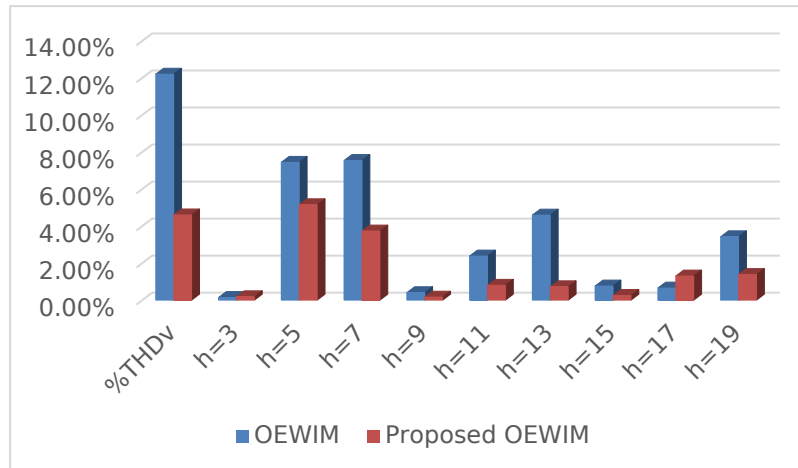


Fig. 8. Comparison between Three-Level OEWM scheme and proposed OEWM configuration

4.1 Experimental Validation - Three -Level OEWM scheme

An 11.5 kVA, two-level inverter with open loop V/f control was used to drive a 5-HP, 415V, 1430 rpm induction motor utilizing the traditional SVPWM-based three-level OEWM system. The Fig. 9 depicts the experimental configuration. The proposed scheme was implemented using the dSpace DS1104 RTI platform with a 20 kHz sampling frequency. The pole voltages of inverter-1 (V_{RO}) and inverter-2 ($V_{R'O'}$) and effective pole voltage ($V_{RO} - V_{R'O'}$) of the three-level OEWM scheme for modulation index $m = 0.8$ is given in Fig. 10. The resultant motor phase voltage and current waveforms of SVPWM-controlled OEWM for modulation index $m = 0.8$ is shown in Fig. 11. The Fast Fourier Transform (FFT) spectrum of motor phase voltage for modulation index, $m = 0.8$ is given in Fig. 12 with a %THDv value of 10.8%. The three-level OEWM scheme has also experimented with different modulation indices $m = 0.2, 0.4, 0.6$ and 1.1 . Motor phase voltage and FFT spectrum for $m=0.6$ shown in Fig. 13 gives a %THDv of 12.77%. Phase voltage and FFT spectrum for $m=0.4$ shown in Fig. 14 gives a %THDv of 14.19%. Fig. 15 gives the FFT analysis for $m=0.2$ with %THDv of 8.27%. For Overmodulation ($m=1.1$) FFT analysis in Fig. 16 shows a %THDv of 12.52%. The distorted voltage leads to non-sinusoidal current injection, which in turn produces torque ripples which causes mechanical vibrations, acoustic noise, shaft fatigue and motor heating.

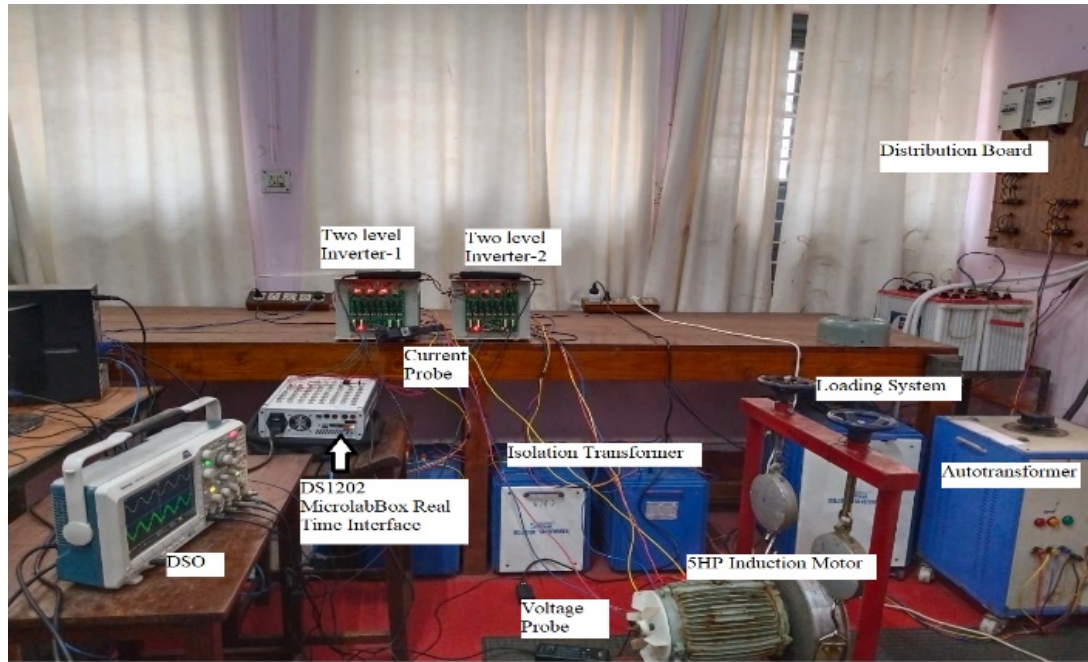


Fig. 9. Hardware setup of the proposed scheme

4.2 Experimental Validation - Proposed OEWIM scheme

The proposed SVPWM based active filter control of dual inverter fed OEWIM scheme was experimented on a 5-HP 415V, 1430 rpm induction motor with spring balance loading. The proposed schemes was implemented using the dSpace DS1104 RTI platform, which has a 20 kHz sampling frequency. Fig. 17 displays the effective pole voltage phase voltage of the proposed OEWIM scheme for $m = 0.8$ as well as the pole voltages of inverters 1 and 2. The resulting motor phase voltage and current waveforms of the suggested OEWIM for modulation index $m = 0.8$ are displayed in Fig. 18. The suggested scheme's motor phase voltage and FFT spectrum with $m = 0.8$ in Fig. 19 display a 5.2% THD. The proposed scheme was also experimented for modulation indices $m = 0.6$, 0.4, 0.2 and 1.1. The motor phase voltage and FFT spectrum of the proposed design with $m = 0.6$, $m = 0.4$, and $m = 0.2$ are shown in Fig. 20, Fig. 21, and Fig. 22 respectively. The motor phase voltage, motor current, and pole voltage of inverters 1 and 2 are shown in Fig. 23, and the FFT analysis for the recommended design with $m = 1.1$ is shown in Fig. 24.

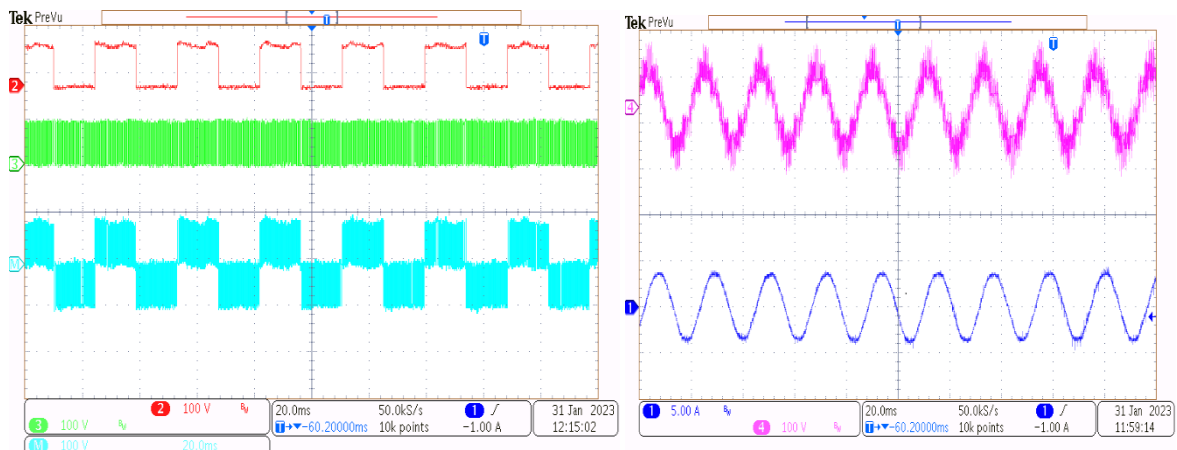


Fig. 10. Pole voltages of inverter - 1 and inverter - 2 and effective pole voltage of the

three-level OEWIM scheme for modulation index $m = 0.8$

Fig. 11. Resultant motor phase voltage and current waveforms of SVPWM-

controlled three-level OEWIM for modulation index $m = 0.8$

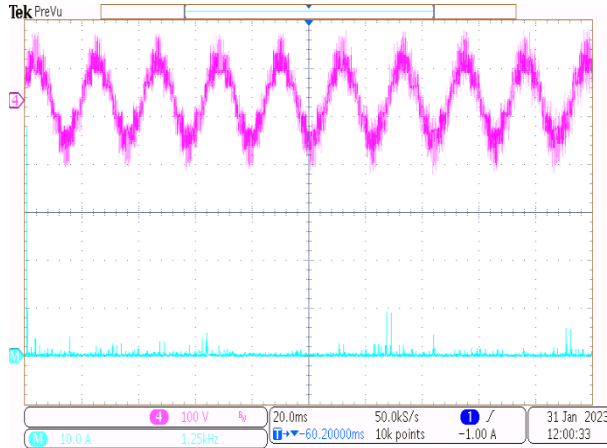


Fig. 12. Fast Fourier Transform (FFT) spectrum of motor phase voltage for modulation index $m = 0.8$

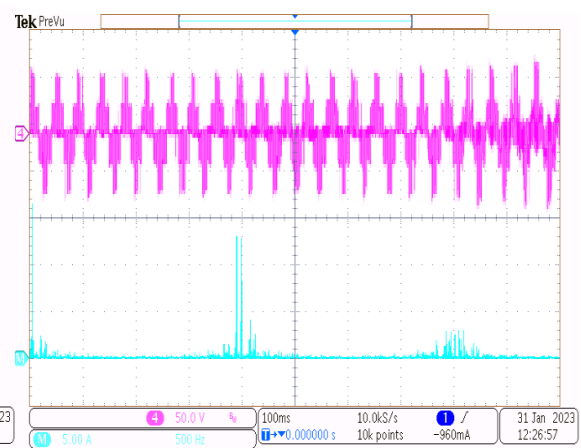


Fig. 13. Motor phase voltage and FFT spectrum for $m=0.6$

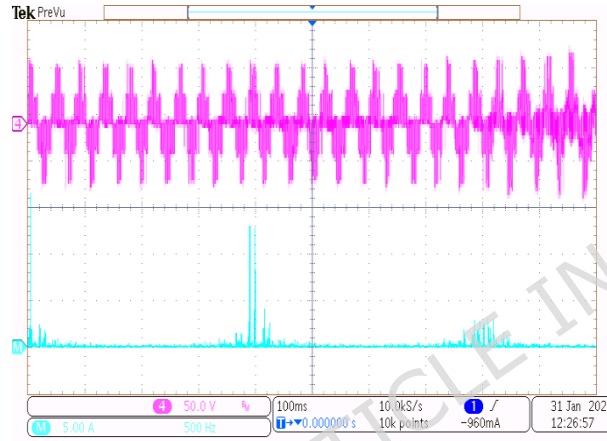


Fig. 14. Motor phase voltage and FFT spectrum for modulation index $m=0.4$

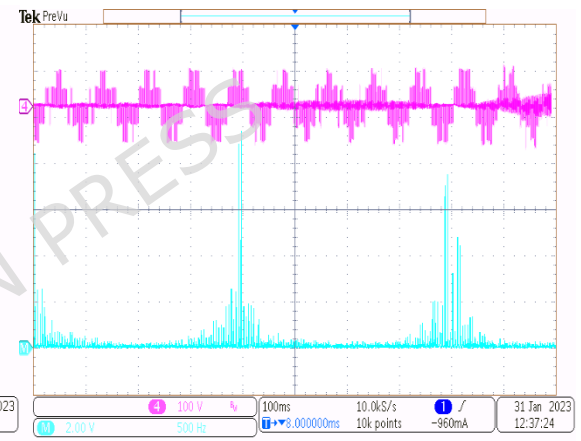


Fig. 15. Motor phase voltage and FFT spectrum for modulation index $m=0.2$

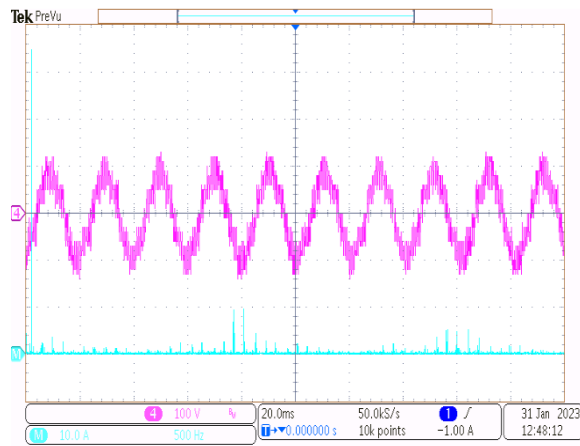


Fig. 16. Motor phase voltage and FFT spectrum for modulation index $m=1.1$

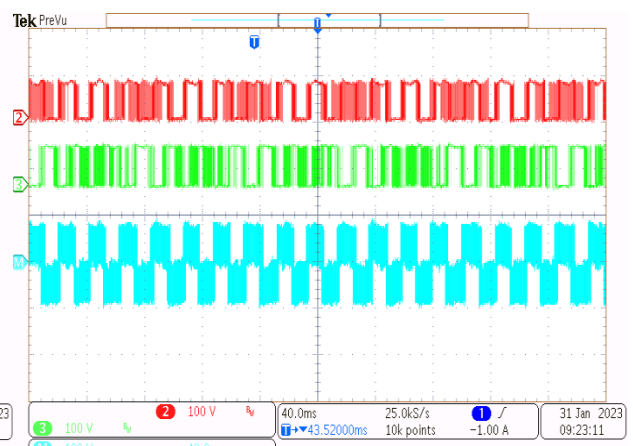


Fig. 17. Pole voltages of both inverters and effective pole voltage of the proposed OEWIM scheme for $m = 0.8$

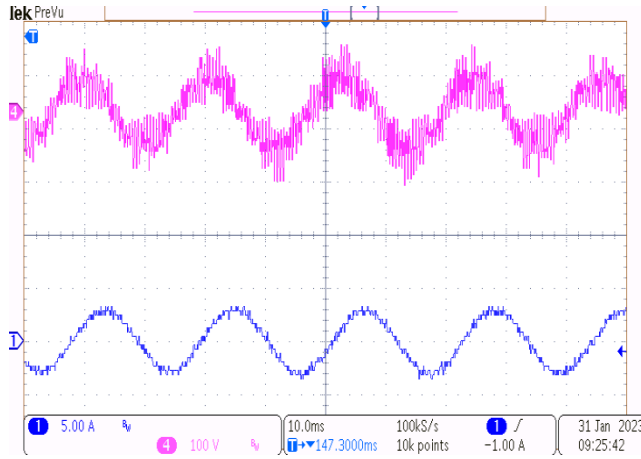


Fig. 18. Resultant motor phase voltage and current waveforms of SVPWM-controlled proposed OEWM for modulation index $m = 0.8$

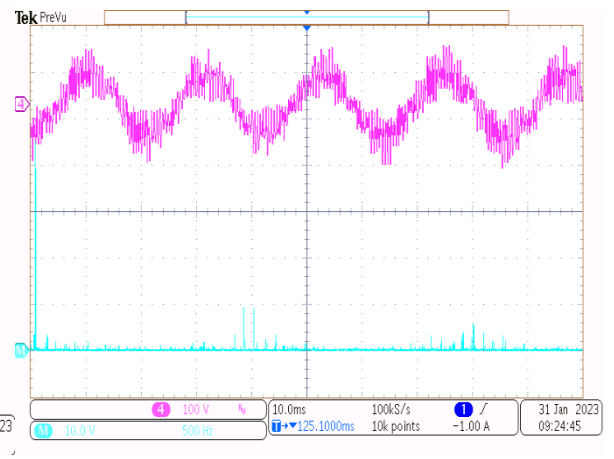


Fig. 19. FFT spectrum and motor phase voltage of the proposed design with modulation index $m = 0.8$

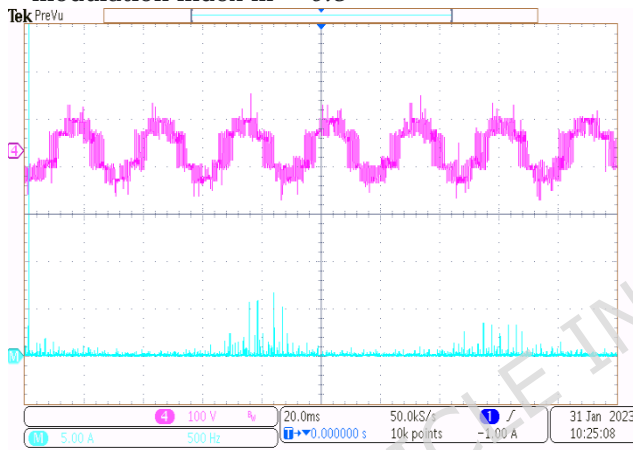


Fig. 20. FFT spectrum and motor phase voltage of the proposed design with modulation index $m = 0.6$

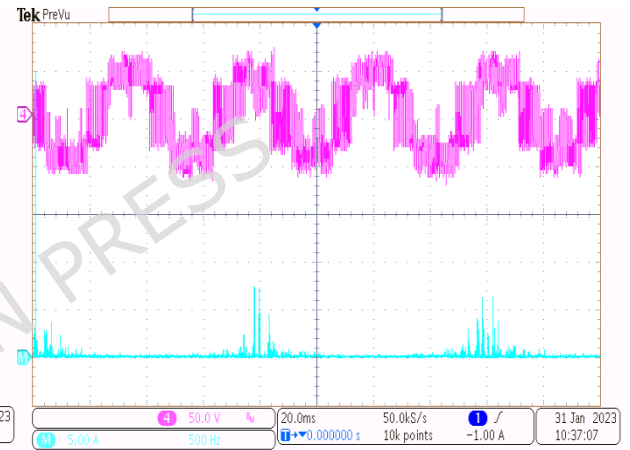


Fig. 21. FFT spectrum and motor phase voltage of the proposed design with modulation index $m = 0.4$

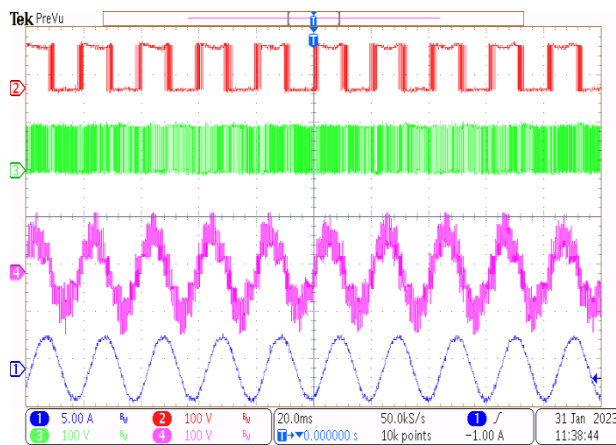


Fig. 22. The proposed scheme's dual inverters pole voltages, motor phase voltage and motor current with $m = 1.1$

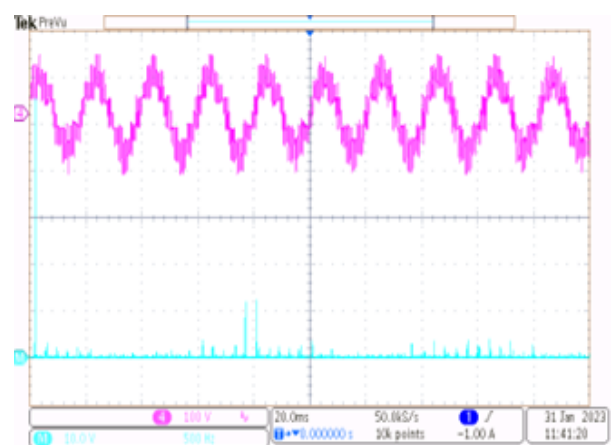


Fig. 23. Motor phase voltage and FFT spectrum of the proposed scheme with $m = 1.1$

	%THD _v					Individual Harmonic %								
	$m=0.8$	$m=0.6$	$m=0.4$	$m=0.3$	$m=1.1$	$h=3$	$h=5$	$h=7$	$h=9$	$h=11$	$h=13$	$h=15$	$h=17$	$h=19$
Neutral Point Clamped Three-Level Inverter	14.2 %	16.23 %	19.32 %	9.50 %	12.52 %	0.18 %	5.15 %	3.73 %	0.16 %	0.80 %	0.72 %	0.25 %	1.30 %	1.38 %
Three-Level inverter with OEWM	10.8 %	12.77 %	14.19 %	8.27 %	12.50 %	0.14 %	7.44 %	7.54 %	0.40 %	2.38 %	4.59 %	0.76 %	0.65 %	3.42 %
Proposed OEWM Scheme	5.2 %	9.40 %	10.02 %	7.20 %	6.50 %	0.06 %	1.16 %	1.00 %	0.05 %	0.18 %	0.21 %	0.07 %	0.11 %	0.34 %

Table 1 Comparison of the proposed OEWM scheme with Three-Level OEWM and NPC schemes

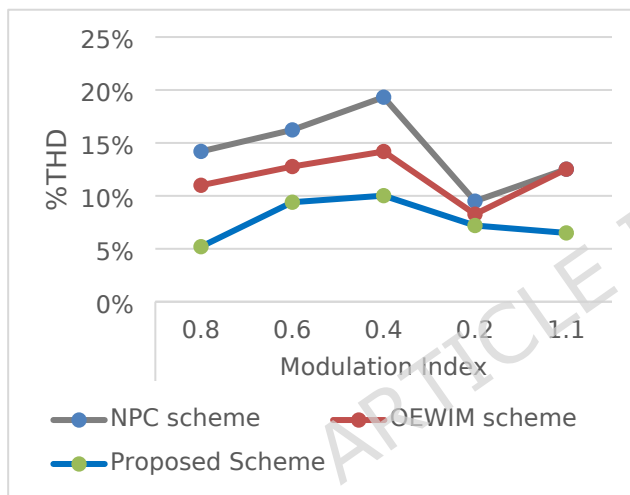


Fig. 24. Performance comparison of the proposed scheme with OEWM and NPC schemes based on %THD

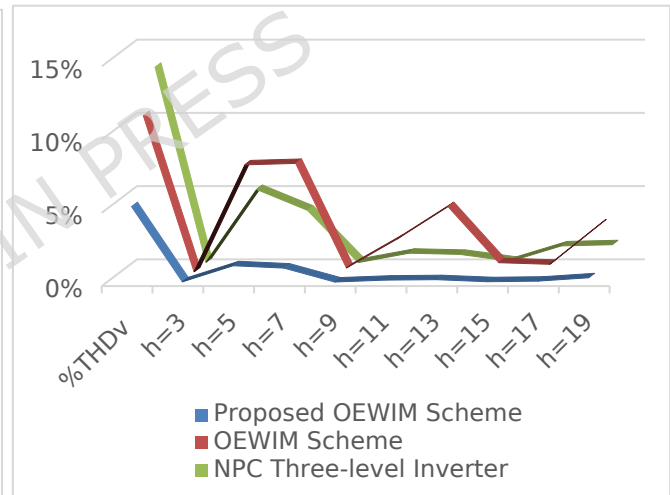


Fig. 25. Comparison of proposed scheme and OEWM and NPC schemes for lower harmonic distortion

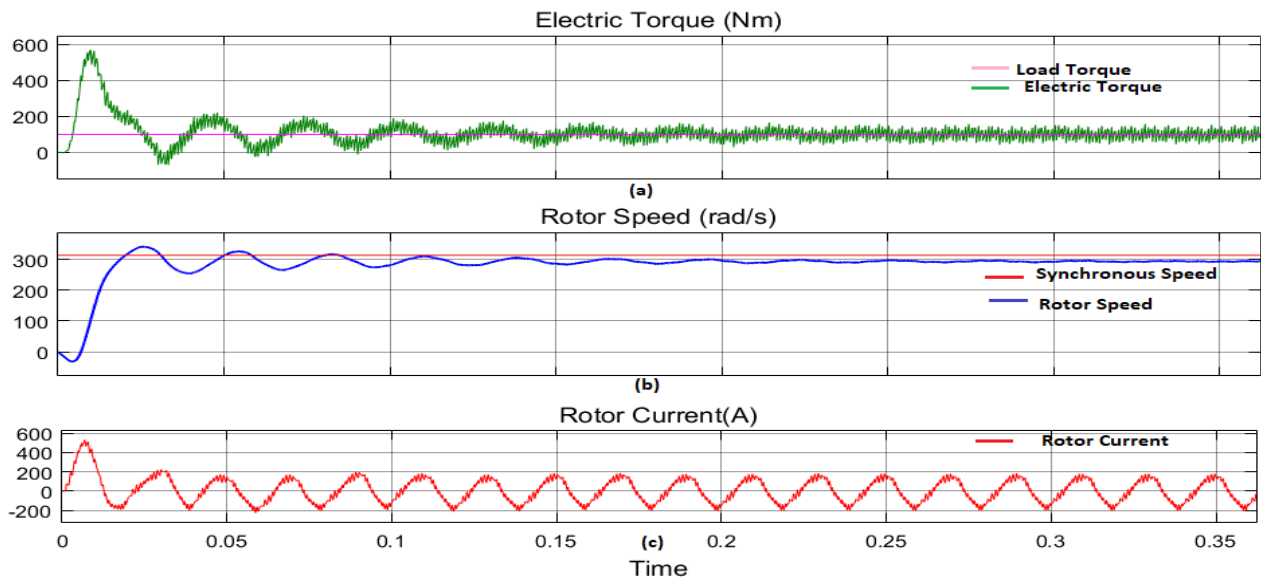


Fig. 26. Dynamic performance of the OEWM machine driven by the proposed power supply and control strategy, (a) Electric Torque (Nm), (b) Rotor Speed and (c) Rotor Current

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5 Comparison with existing schemes

The proposed active filter based OEWIM scheme is contrasted with the traditional open-end winding based three level inverter and Neutral Point Clamped inverter based on percentage THD with modulation indices, $m = 0.8, 0.6, 0.4, 0.2, 1.1$. Also, individual lower order harmonic distortions in voltage waveforms of the three schemes were compared. Table 1 presents the total harmonic distortion of voltage (%THDv) for different modulation indices and the individual harmonic components of the inverter output. The Neutral Point Clamped three-level inverter, the three-level inverter with OEWIM, and the proposed OEWIM scheme were simulated using the same machine model, load conditions, and switching frequency to ensure a fair comparison.

The results show that the proposed OEWIM scheme significantly reduces %THDv across all modulation indices and achieves substantial suppression of dominant low and medium order harmonics compared to the other two configurations. Minor variations in specific higher-order harmonics are observed but remain at very low magnitudes and do not affect the overall performance of the drive. The comparison between the proposed system and the traditional techniques for reduced harmonic distortion is displayed in Fig. 25 and Fig. 26. The results confirm that the suggested strategy improves drive performance by outperforming traditional schemes based on lower harmonic distortion.

Fig. 26 illustrates the dynamic and steady-state performance of the OEWIM drive supplied by the proposed power supply and control scheme. The electromagnetic torque response in Fig. 26 (a) shows a noticeable reduction in torque ripple during both transient and steady-state conditions, which contributes to improved mechanical performance and reduced stress on the drive system. The start-up response shown in Fig. 26 (b) demonstrates smooth acceleration from standstill to the reference speed without overshoot or instability, confirming the effectiveness of the proposed scheme under variable frequency operation. The stator current waveform in Fig. 26 (c) exhibits reduced distortion and ripple, indicating improved current quality compared to conventional inverter-fed OEWIM configurations.

The steady-state harmonic spectrum of waveforms confirms significant suppression of dominant low- and medium-order harmonics, consistent with the quantitative results reported in Table 1 and Figure 8. These results validate that the proposed scheme enhances torque quality and harmonic performance while maintaining stable operation during dynamic conditions such as start-up.

6 Conclusion

To mitigate the motor's phase voltage harmonic distortions, a dual two-level inverter-fed OEWIM drive is suggested. In the proposed scheme when compared to the standard open-end winding induction motor drive, a primary two-level inverter provides the motor with the necessary power and a secondary inverter is employed to reduce harmonic distortions in the voltage waveforms. The secondary two-level inverter uses a series filter control approach, and the primary inverter is controlled using space vector pulse width modulation. The fundamental component of the primary inverter's output voltage is measured in order to provide the reference voltage for the filter scheme. The active filtering control method eliminates harmonic in the motor waveforms with minimal computational steps. It will enhance drive efficiency and reduces torque ripple as phase-voltage harmonic distortions decreases. The suggested approach is simulated and contrasted with an SVPWM controlled OEWIM drive and a three-level neutral point clamped inverter. The findings shows that the proposed OEWIM scheme has minimal phase-voltage distortions than the three-level OEWIM systems for all modulation indices. The experimental results also confirms that the proposed scheme is superior over the OEWIM scheme without any add-on hardware requirement.

Conflict of interest

On behalf of all authors, the corresponding author states that there is no conflict of interest

Data availability

All data generated or analyzed during this study are included in this published article.

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